

Guaranteed-Companion Seating: Current Implementation and Evidence Guide

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This report specifies the visual-simulation implementation after the follow-up audit. The formal manuscript, including proofs, related work, figures, experimental results, and limitations, is `article/article.tex`. Historical proposal documents describing maximization of two-friend coverage are superseded. The original external prototype and historical collusion experiment files are unavailable; this repository implements and tests the stated mechanism independently.

1 Inputs and validity

Each participant has a directed submitted list. In the defended reference case, a list has at least four distinct non-self names, at least two from the participant's grade, or is empty. The study generator and attack constructions use at most eight names. The generic runner does not enforce that cap for arbitrary custom inputs; the privacy calculation states it as an observer assumption. The undefended scenario deliberately permits one-name lists. A generated honest list failing the minimum rules becomes a recorded nonsubmission, rather than being padded with invented names. This policy is a simulation assumption and does not establish fairness for real participants.

For each rotation, every participant must appear exactly once, tables must have their exact capacities, and all grade restrictions must hold. Every participant whose submitted list is nonempty must share a table with an eligible listed peer. Filtering out cross-grade names does not turn an unsatisfiable submitter into a nonsubmitter. An empty effective list remains an error.

2 Admission screens and their limits

The screens operate on the effective graph separately for mixed and same-grade rotations. Successor closures of at most 64 vertices are checked for a split into guarantee-closed parts. A set is guarantee-closed when each obligated member has at least one eligible name within it. Self-containment, required by closure forcing proofs, means all such outgoing names stay inside.

Targeted candidates additionally include strongly connected cores of sizes 2–12 in the original graph and every graph obtained by deleting a single vertex. Candidates are deduplicated, sorted by size and identifiers, and limited to 256. Each targeted relaxation restores every original outgoing edge of the core, adds all its outside names, and removes only the outside vertices' obligations. If even this easier model cannot separate core members into two supported parts, every valid original chart keeps the core together. A found relaxed split is not a global feasibility witness. Outside anchors are not returned merely because they appear in the certificate.

Relaxations of at most 20 vertices use exact enumeration; larger ones use CP-SAT with ten deterministic-time units, one worker, and seed zero. Results distinguish proved coercive, proved separable under the test, and unresolved. UNKNOWN is unresolved and does not imply coercion. A degree-three shortcut counts distinct neighbors and applies only when every tested vertex has an obligation. Candidate counts and limits are recorded, so passage is not represented as complete attack resistance.

Table 1: Reference configuration. Complete generator, submission, and solver configurations are retained in every trace.

Parameter	Value
Population	257 (132 grade 11; 125 grade 12)
Tables	17×7 and 23×6
Rotations	16, alternating mixed and same grade
Defended study lists	None or 4–8 names; at least 2 same grade
Reference nonsubmitters	0
Popularity / within-grade weight	$\sigma = 0.6 / 9$
Reciprocity target / realized	0.5 / 0.490
Realized within-grade fraction	0.904
Reference communities	$\mu = 0, \omega = 0$
Annealing budget / temperature	300,000 iterations / $25 \rightarrow 0.2$ (integer energy)
CP budget / workers	3.5 deterministic-time units / 8
Pre-feasibility budget	20.0 deterministic-time units per state
Runtime versions	Python 3.14.7, NumPy 2.5.2, OR-Tools 9.15.6755

The demand test handles a self-contained core containing n_0 nonsubmitters and n_1 submitters, together with submitting followers whose nonempty eligible lists stay inside the core. At most $n_0 + \lfloor n_1/2 \rfloor$ tables can host them. Compare their population with that many largest capacities in the admissible grade pool. The senior regression has demand 42 and only 41 available seats; it must not borrow a junior seven-seat table. Missing capacity-pool information produces an unresolved check.

A screen-enabled year requires review if the input contains returns, unresolved checks, invalid lists, or submitters without eligible names. The program does not manufacture revised preferences. The screened demonstration explicitly supplies one omission-star resubmission after returning a six-person stratified attack. Shared-anchor charts intentionally disable enforcement while saving a diagnostic of the revised screen.

3 Separate feasibility check

A hard-constraint CP-SAT model is attempted for both distinct eligibility states before year generation. FEASIBLE or OPTIMAL is a positive certificate; INFEASIBLE is a negative certificate; UNKNOWN establishes neither. A later validated chart can supply a positive witness after an inconclusive precheck. If the complete pipeline cannot produce a valid chart, it raises an unsuccessful outcome instead of publishing an invalid chart.

4 Objectives and solver stages

Let s_i be a submitter’s eligible listed-peer count in the current table, ν the number with $s_i = 0$, and m_{ab}, α_{ab} disjoint historical counters. A meeting increments m if neither participant listed the other and α otherwise. Annealing uses

$$C = \sum_i \max(0, s_i - 1) + 0.1 \sum_{a < b \text{ together}} (5m_{ab} + 2\alpha_{ab}), \quad E = 10C + 10000\nu.$$

The unscaled energy is $C + 1000\nu$, with temperature $2.5 \rightarrow 0.02$; the implementation scales both energy and temperature by ten, using $25 \rightarrow 0.2$. These equivalent forms must not be mixed when describing acceptance probabilities.

Construction. `pod_greedy` prioritizes constrained participants, forms pods, and packs them into the physical tables. Packing fallback can create unsupported singletons. Construction is not a feasibility certificate.

Repair. `swap_repair` uses guarded swaps and compound moves that reduce the unsatisfied set without orphaning previously satisfied submitters. State and derived values must be restored after a rejected move. Repair may stop before feasibility.

Annealing. Proposals mix uniform, directed, and group moves. Two-table group moves close supporting dependencies, preserve capacities, and retry padded closure. Accepted proposals cannot increase ν . Once feasible, the chain stays feasible; before feasibility, identities of unsatisfied participants can change at constant ν . Metropolis acceptance operates on the exact affected-table cost delta. The best chart is saved lexicographically by violations and cost. After an infeasible initial run of I iterations, up to six repair-and-anneal retries use $\max(I/2, 20000)$ iterations each.

CP improvement. Assignment variables have exact-one and capacity constraints. Supporting indicators obey $y_{it} \leq x_{it}$, $y_{it} \leq \sum_{j \in L_i} x_{jt}$, and $\sum_t y_{it} \geq 1$. A forced two-plus indicator z_i satisfies

$$\sum_{j \in L_i} x_{jt} \leq 1 + (|L_i| - 1)z_i + |L_i|(1 - x_{it}).$$

The CP surrogate minimizes $300 \sum_i z_i + 100 \sum_{\{a,b\} \in P_{r-1}} (5m_{ab} + 2\alpha_{ab})q_{ab}$, where P_{r-1} contains only the previous rotation’s pairs and q_{ab} is forced on if that pair meets again. Coefficients use full history. The model receives consistent assignment and auxiliary hints, which need not themselves be feasible. A feasible CP candidate is accepted only if it does not worsen the incumbent’s full annealing energy. A remaining invalid chart triggers a final CP attempt with budget $6 \max(b, 1)$ for initial CP budget b . Only a final validated chart is released.

5 Metrics and baseline

Coverage and exactly-one percentages use submitters as the denominator. Distinct-peer means use all participants. A repeated pair is an unordered pair sharing a table now that also met earlier; its count differs from the weighted historical cost. Coalition member-weighted cluster size is $\sum_G |G \cap C|^2 / |C|$, using six or seven members as appropriate. All members sharing a table is distinct from filling it completely.

The random baseline uniformly permutes participants into identical table slots with the same eligibility states. Its pairwise meeting probability in a block of size n is $\sum_t c_t(c_t - 1) / (n(n - 1))$. Expected distinct contacts sum $1 - \prod_r (1 - q_{ab,r})$ over peers. Random companion coverage uses the hypergeometric probability, weighted by table seat share. The applicable reference expectation is 72.29 distinct peers for the alternating schedule, not the historical all-mixed value 74.72.

The exact privacy probe treats each observed tablemate set as a hitting constraint and uses a public upper list cap. Excluding a name forces that name only if the minimum remaining hitting set exceeds the cap. Private truth is used afterward to score correctness. Full true-positive recovery of a short list and unique identification under the public cap are separate fields. The reference has 613 forced entries out of 2056 (29.8%). No privacy guarantee is claimed.

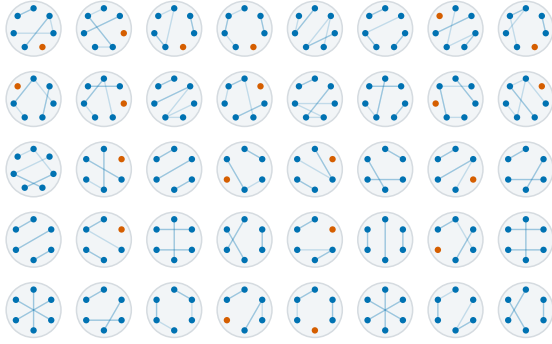
6 Evidence, reproduction, and website

The experiment ledger records a run before executing it. Terminal statuses retain interruptions, input failures, review requirements, source changes, and other unsuccessful outcomes. A success includes a full trace, input and configuration identity, effective source-file hashes, full Git identity, dirty status, runtime versions, and a trace checksum. Resumption requires matching identity and valid evidence; old summary-only results are not promoted to verified traces.

The fresh study’s inventory is in `results/verified_experiments.json`; exact commands and concurrency conditions are in `results/STUDY.md`. The executable source used by the study is preserved by effective hash under `results/source_snapshots/`. Independently written verification

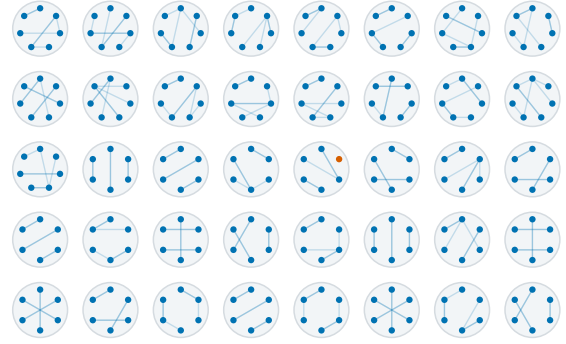
Honest population seed 2, rotation 1 (mixed): every table and student

a Pod construction



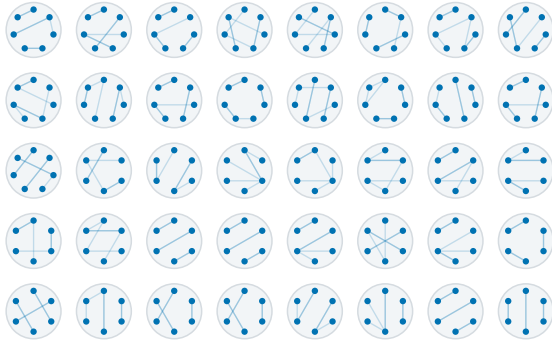
Without a listed peer: 20
Extra listed peers: 19

b Swap repair



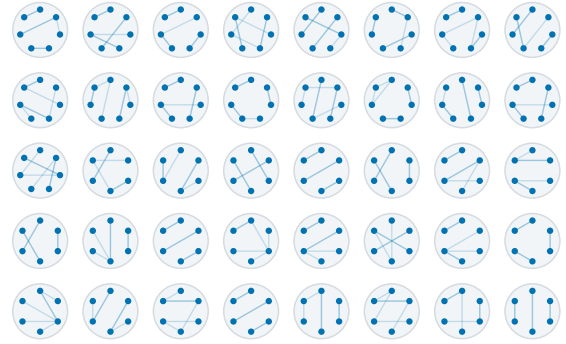
Without a listed peer: 1
Extra listed peers: 16

c Annealing



Without a listed peer: 0
Extra listed peers: 0

d Final accepted seating



Without a listed peer: 0
Extra listed peers: 0

Blue: has a listed peer Orange: lacks a listed peer Gray: no submission

Figure 1: Actual construction, repair, annealing, and final endpoints from one verified rotation. The early stages can violate the companion constraint. These are saved assignments, not fabricated illustrations of a browser optimizer.

scripts recompute all final and intermediate chart metrics and check privacy implications with a separate CP formulation. The manuscript tables and figures read this retained evidence.

The static website replays six seed-seven scenarios and four actual solver endpoints for each honest rotation. It displays intermediate violations, the accepted final phase, current eligibility, dynamic coalition size, and measured outcomes separately from structural statements. It does not optimize arbitrary user inputs. The updated manuscript is downloaded from `docs/article.pdf`.

```
python -m pip install -r requirements-paper.txt
python -m pytest -q
python paper/assemble_evidence.py --retained-source
python paper/make_tables.py
python paper/make_figures.py
cd article
latexmk -pdf -interaction=nonstopmode article.tex
```

7 Scope of conclusions

The implementation validates every retained chart and supplies sound certificates for examined coercive cores. It does not establish complete anti-capture screening, universal feasibility, optimal full-year schedules, empirically calibrated network behavior, fairness, privacy preservation, or improved wellbeing. The omission star permits $3 + 3$, $4 + 2$, and 6 ; its bound of three does not force two triples. The cycle-extension reasoning is prior work, attributed in the formal article to Thomassen and Alon’s discussion of school coalitions. A real pilot, author declarations, and journal acceptance require evidence or decisions outside these synthetic experiments.